

RE-ENTERING UNFINISHED MUSIC: SESSION-CONDITIONED RETRIEVAL OF PERSONAL AUDIO FRAGMENTS WITH REVERSIBLE DAW ACTIONS

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ABSTRACT

Musicians accumulate large libraries of unfinished audio fragments, yet re-entering an unfinished project offers little support for deciding what to try next. We formulate *session-conditioned continuation retrieval*: given the live state of a DAW project, rank a musician’s own fragments by their utility for the next creative step. Pocket Producer implements this in Audiotool: it derives a transparent fingerprint from the live project, ranks personal fragments with frozen audio–text representations, and inserts any recommendation through a single reversible transaction. On 1,755 weak pairs from the Freesound Loop Dataset, frozen CLAP cosine reaches 0.883 pairwise accuracy and learned rerankers (up to 335K parameters) cannot beat it; a five-parameter fusion of cosine, tempo, key, and tag signals reaches 0.794 on the hardest negatives, +4.0 pp over cosine (0.754; one-sided sign test $p = 0.031$). Session-structure signals (instrument role gap, harmonic compatibility) show no gain on pack-membership labels, and an external evaluation (9 raters, 20 pairs, Fleiss’ $\kappa = 0.294$) shows continuation preference is subjective but not random—together motivating task-specific evaluation beyond acoustic similarity.

1. INTRODUCTION

When a musician returns to an unfinished loop-based project, the central question is not “what sounds similar?” but “what would be useful as the next step?” Existing work on loop compatibility [1–3] and stem coherence [4, 5] asks whether two elements can co-exist in a mix; our query is a *live session state*, and the candidates are the musician’s own creative memory, not a public catalog.

We define **continuation utility** as the property that a candidate fragment helps the musician make progress in the current session—subsuming tempo fit, timbral contrast, structural gap-filling, and stated intent. We contribute:

1. **Task**: session-conditioned ranking of personal fragments with a preregistered primary metric (pairwise



Figure 1. Pocket Producer retrieval loop: live session fingerprint $\rightarrow$ ranking $\rightarrow$ reversible insertion $\rightarrow$ preference feedback.

accuracy over human continuation preferences).

2. **System**: a working prototype in a real DAW (Audiotool) with live document reading, reversible insertion, and explainable evidence.
3. **Protocol**: a backbone bake-off, a baseline comparison (learned rerankers lose to frozen cosine; multi-signal fusion gains +4.0 pp on hard negatives), and a blind A/B evaluation with 9 external raters.

2. POCKET PRODUCER

Figure 1 shows the loop; the system reads and writes Audiotool projects through the Nexus SDK.

2.1 Session Fingerprint

When a project is opened, the live Nexus document is mapped deterministically to a `SessionFingerprint`: tempo, track counts and classified track types, playhead position, and a user-editable text intent field. Only this summary leaves the browser; missing fields are never imputed. Session audio regions are encoded with MS-CLAP 2023 [6] into a mean session embedding (CLAP’s joint audio–text space also serves text intent); per-region instrument roles and 12-dim CQT chroma are extracted for the session-structure signals below.

2.2 Continuation Utility Ranking

Candidate fragments are scored in two stages. First, per-space retrieval runs independently over four signal spaces: (1) structured metadata (tempo/key/tag matching), (2) frozen CLAP embeddings (cosine similarity), (3) *role-gap fill*—a zero-shot instrument probe boosts candidates whose predicted role fills a gap in the session, a compositional signal cosine similarity cannot capture, and (4) *harmonic compatibility*—transpose-invariant



Table 1. Backbone bake-off on 2,558 FSLD items. R@10 = same-source Recall@10; Genre/Role = linear-probe accuracy.

Backbone	Dims	R@10	Genre	Role	Bytes/frag
laion-clap-music	512	0.713	0.415	0.556	2,048
msclap-2023	1024	0.692	0.384	0.533	4,096

Table 2. Baseline comparison on FSLD weak pairs (LAION-CLAP 512-dim, 5 seeds). Hard-sim = hard_similar negatives; Hard-tk = hard_tempo_key. Bold = best per column.

Model	Params	Overall	Hard-sim	Hard-tk
cosine-mean	0	0.883	0.754	0.911
deepsets-v1	164K	0.735	—	—
pocketrank-v1	335K	0.742	—	—
cosine+session	2	0.882	0.749	0.911
fusion (5-signal)	5	0.894	0.794	0.897
fusion (7-signal)	7	0.892	0.792	0.894

cross-correlation of CQT chroma vectors scores tonal fit across all 12 semitone shifts (65.2% vs. 50% chance pack-membership prediction on 500 FSLD pairs). Results are fused with Reciprocal Rank Fusion [7]; a learned reranker ($\leq 500K$ parameters, CPU-only) can be applied on top, and a deterministic rule baseline with structured evidence codes serves as an always-available fallback.

2.3 Reversible Audiotool Interaction

A recommendation can be previewed and inserted in a single Nexus transaction; an `InsertReceipt` records every created entity ID so undo restores the exact prior state (verified with the official WASM validator), and all interactions are logged as preference feedback.

3. PRELIMINARY EVALUATION

3.1 Backbone Bake-off

On 2,558 items from the Freesound Loop Dataset (FSLD) [8], LAION-CLAP Music [9] (512-dim) outperforms MS-CLAP 2023 (1024-dim) on all three probes (Table 1) at half the storage per fragment. We nevertheless deploy MS-CLAP 2023 because its training-data license audit is complete—representation quality and deployment eligibility are separate decisions.

3.2 Baseline Comparison and Multi-Signal Fusion

We constructed 1,755 leave-one-out pairs from FSLD, using same-pack membership as a weak proxy for continuation utility, with three negative types: *easy* (tempo or key incompatible), *hard_tempo_key* (tempo and key both compatible), and *hard_similar* (highest embedding cosine from another pack). Splits are grouped by source pack; Table 2 reports 5-seed mean validation pairwise accuracy (val fraction 0.25).

Frozen cosine similarity (0.883) is already a strong baseline: two learned rerankers with up to 335K param-

eters trained with BPR loss [10] score below it (0.735, 0.742), overfitting to noise in the weak labels. However, a five-parameter logistic fusion of audio cosine (mean and max pooling), tempo match, key match, and tag Jaccard overlap—trained with the same BPR loss—reaches **0.794 on hard_similar negatives**, a +4.0pp gain over cosine alone (0.754); the gain holds in all 5 seeds (one-sided sign test $p = 0.031$). These are embedding-similar items from different packs where cosine is uninformative; the gain comes from music-theory signals the embedding cannot encode. On *hard_tempo_key* negatives, where tempo and key features are neutral by construction, fusion (0.897) stays close to cosine (0.911), ruling out label leakage.

Adding the session-structure signals (Section 2.2) does not improve the fusion: 7-signal results (0.892 overall, 0.792 *hard_similar*) are indistinguishable from 5-signal, and *cosine+session* (0.749) sits slightly below cosine alone. This is expected: pack membership reflects recording context, not whether a candidate fills a compositional gap—purpose-built evaluation is needed for session-conditioned signals.

3.3 Author Pilot Evaluation

Under a preregistered protocol (frozen 2026-07-30), the author labeled 60 blind A/B pairs; 42 received a preference and 18 were “neither” (30%). No ranker exceeds 0.571 pairwise accuracy (chance=0.5), and cosine-agreement is 57.1% (24/42)—statistically indistinguishable from chance (Wilson 95% CI [0.42, 0.71]). Agreement rises monotonically with annotator confidence, and the 30% “neither” rate is a judgment a similarity ranking cannot express. Exploratorily, the role-gap signal fires on only 7 of 42 pairs but agrees with the human preference in 5 of those 7, suggesting a sparse but precise structural signal. Author labels are excluded from the evaluation set per the preregistration.

3.4 External Blind Evaluation

Nine external participants labeled the same 20 blind A/B pairs (180 labels) with a five-point confidence scale. Fleiss’ $\kappa = 0.294$ indicates fair inter-rater agreement—typical for subjective music-preference tasks [11]—with 10 of 20 pairs reaching $\geq 7/9$ majority and two unanimous (9/9). Most pairs elicit a clear preference (only 8.9% “neither” versus the pilot’s 30%), suggesting context information aids decision-making; these are descriptive statistics with no population-level significance claims.

4. DISCUSSION AND LIMITATIONS

Weak labels $\neq$ ground truth. Weak labels cannot establish continuation utility, and the session fingerprint still lacks temporal structure and rhythmic analysis; purpose-built human evaluation remains necessary.

Scope and privacy. We measure perceived relevance, not creative output quality; session audio is never stored and OAuth tokens never leave the browser.

155 **Acknowledgements.** The author thanks the Audiotool 204
156 team for the Nexus SDK and the FSLD creators [8] for 205
157 the publicly available dataset. 206

158 **AI Usage Statement.** Claude Code (Anthropic) was used 207
159 for engineering implementation and documentation draft- 208
160 ing. All research claims, experimental design, analysis, ci- 209
161 tations, and figures are the sole responsibility of the author, 210
162 who has verified each element against the linked artifacts 211
163 and source publications. 212
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